

particular sensor. The remaining five bits are manipulated to designate two different addresses to two different data nibbles. These five bits, once set, must remain set forever. So, during scanning of sensors, although the logic levels of A-B-C scanning input lines of the ADC go on changing sequentially from 000 to 111 in cyclic order, the said five other bits of the address word remain fixed to their preset value. As a result, different sensor output is allotted with different 8-bit addresses to both nibbles. This is true for all eight sensors.

At the receiving end, after detection, sequentially-received two pulse trains for two nibbles are fed to two special decoders. Like encoders, decoders, too, have pre-settable eight address lines. If at any time, the preset address of a decoder matches with the instantaneous address of an encoder, the decoder decodes the received signal and extracts the 4-bit data sent for it, and consequently places the same on its four output lines.

The 8-bit output data (4-bit plus 4-bit) from the two decoders are fed to an 8-bit DAC. The DAC in return generates an equivalent analogue current. This current, when allowed to flow through a resistor, develops an analogue voltage (negative with respect to ground) across it. This analogue output of the DAC is measured with the help of a digital multimeter.

Circuit and working

Circuit of the wireless addressable digital thermometer is divided into three parts: transmitter, receiver and power supplies.

Transmitter: Fig. 2 shows the circuit diagram of the transmitter unit for the wireless addressable digital thermometer. Eight LM35 IC temperature sensors (IC9-IC16) are connected to IN0-IN7 inputs (pins 26 to 28 and 5 to 1) of ADC 0808 (IC3). Although the ADC is capable of accepting a total number of eight sensors through its eight input lines, less number of sensors could be used as well as,

PARTS LIST

Semiconductors:	
IC1	- 7404 TTL hex inverter
IC2	- NE555 timer
IC3	- ADC0808, 8-bit ADC
IC4, IC5	- HT12E 2 ¹² series encoder
IC6	- CD4029 CMOS up/down counter
IC7	- 7408 quad 2-input TTL AND gate
IC8, IC17	- 7805, +5V voltage regulator
IC9-IC16	- LM35 temperature sensor
IC18	- 7905, -5V voltage regulator
IC19	- 4011 quad 2-input NAND gate
IC20, IC21	- HT12D 2 ¹² series decoder
IC22	- DAC0808, 8-bit DAC
T1, T2	- BC547 npn transistor
D1, D2	- 1N4007 rectifier diode
D3, D4	- 1N4148 signal diode
BR1	- DB107 bridge rectifier
LED1-LED14	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2, R10-R12, R15, R18, R19, R27-R34	- 1-kilo-ohm
R3	- 2.2-kilo-ohm
R4, R20-R23	- 100-kilo-ohm
R5-R7, R13, R14	- 10-kilo-ohm
R8, R9	- 680-kilo-ohm
R16, R17	- 33-kilo-ohm
R24	- 2.7-kilo-ohm
R25	- 1.2-kilo-ohm
R26	- 5-kilo-ohm
VR1	- 10-kilo-ohm 40-turn trim potentiometer
Capacitors:	
C1	- 500pF ceramic disk
C2	- 1µF, 16V electrolytic
C3	- 10nF ceramic disk
C4, C6, C7	- 1000µF, 25V electrolytic
C5, C8, C9	- 10µF, 16V electrolytic
C10	- 100nF ceramic disk
C11	- 100µF, 16V electrolytic
Miscellaneous:	
X1, X2	- 230V AC primary to 12V-0-12V, 500mA secondary transformer
TX1	- 433MHz transmitter module
RX1	- 433MHz receiver module
TWS1	- Thumbwheel rotary switch
DM1	- 3 1/2-digit digital voltmeter
CON1, CON2	- 3-pin connector
CON3	- 5-pin connector
CON4	- 2-pin terminal connector
ANT.1, ANT.2	- 35mm length wire

whenever desired. IC 7404 (IC1) configured as a CMOS oscillator with the help of resistors R1 and R2, and capacitor C1 feeds the ADC with necessary clock pulses required for conversion processes.

Output voltage of LM35 series IC temperature sensors follows linearly (@10mV/°C) the centigrade temperature of its surroundings, taking 0mV at 0°C temperature. The ADC continuously scans its eight input lines. The scanning process is

governed by a 3-bit binary up counter built around CD4029 (IC6).

The counter places a continuously-changing 3-bit binary number on A-B-C input lines of the ADC (pins 25, 24 and 23). Scanning rate is dependent upon the clock constructed around timer NE555 (IC2), and is 8Hz, approximately. Hence, each of the eight sensors is allowed to send data to the ADC for approximately one-eighth of a second, irrespective of whether all sensors are connected or not.

Here, IC3 is configured in continuous operational mode. So, whenever a particular sensor is addressed, output lines of the ADC reflect the present analogue output status of the sensor. Output of the ADC goes to data input lines of special encoders HT12E (IC4 and IC5); higher nibble to IC4 and lower nibble to IC5, respectively.

As $\overline{TE}$ input (pin 14) of encoders is permanently grounded (logic 0), the encoders are configured to produce encoded data continuously. Data is available at pin 17 of IC4 and IC5. These two encoded digital outputs are alternately steered to TX1 (TX-433MHz), a UHF RF transmitter module, to modulate UHF carrier wave generated by the module. Selection of encoder output is done in the following manner:

Whenever IC2 output pulse goes high, pin 10 of AND gate N3 of IC7 (a quad 2-input TTL AND gate) also goes high, allowing output of IC5 to be steered to TX1 through diode D4. At the same time, due to the presence of transistor inverter T1, pin 13 of gate N4 of IC7 goes low, inhibiting output of IC4 to reach TX1 through gate N4. As soon as the clock pulse returns to logic 0, output of IC4 gets its passage to TX1 through gate N4 of IC7.

So, in essence, analogue data of a sensor is converted and the resultant 8-bit digital data is sent to the remote end using ASK/OOK modulation, in a complete clock cycle of IC2. Modulated signal is radiated into space through a wire (approximately 35cm long), acting as an antenna, connected